

PE semi-auto testing program

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- Popup box immediately after testing done. [Click](#).





Purpose

- Make the testing easier
- Reduce the test workman
- Reduce testing time
- To avoid electric arc that cause by the surface of electrical probe and measure point didn't completely contact together
- Improve programming skill and hardware communication skill

Content

Hardware

Program

How to use ?

Additional

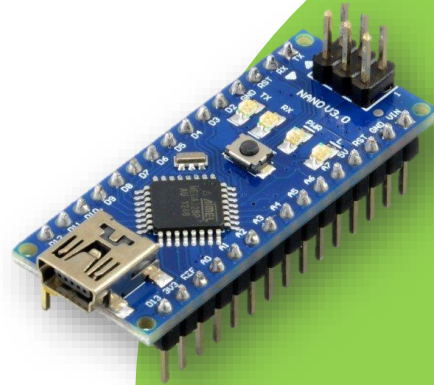
Result

HARDWARE

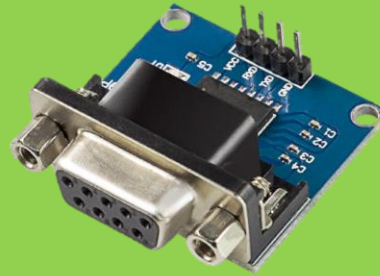
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Component

Controller



RS232 to TTL



Foot switch



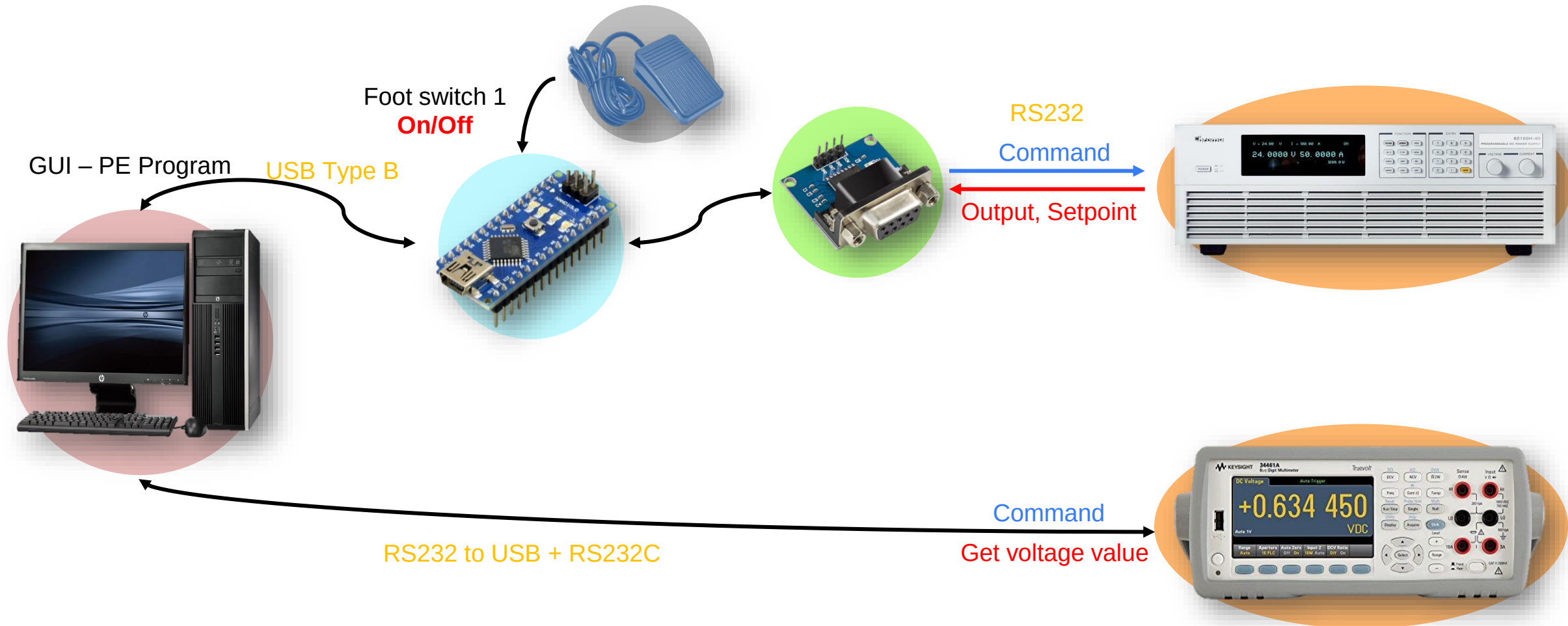
DC-Source



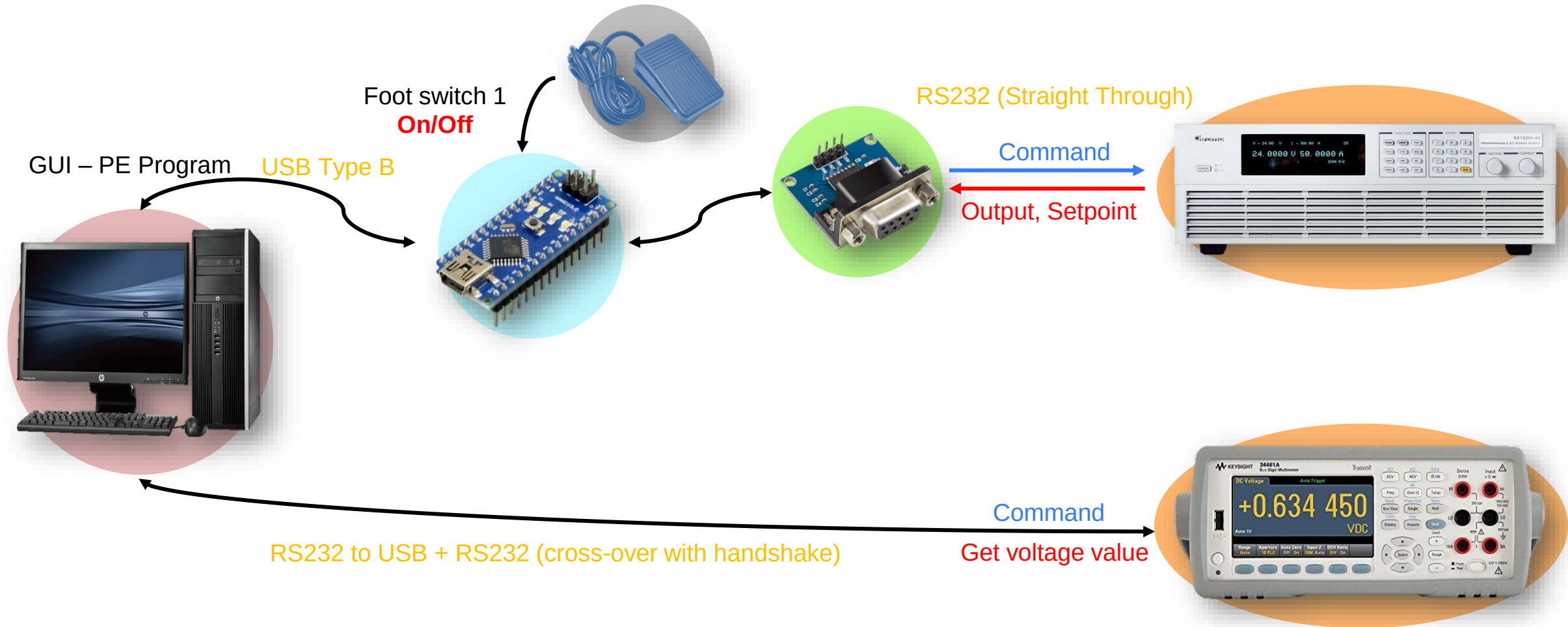
Digital Multimeter



Overview

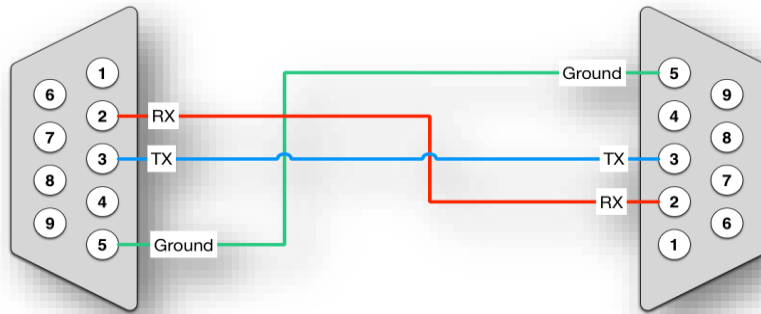


Overview

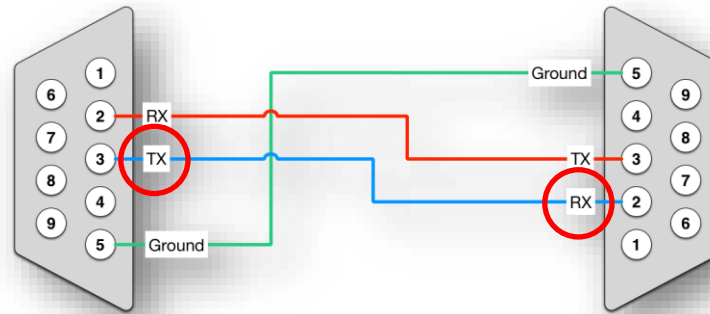


Information

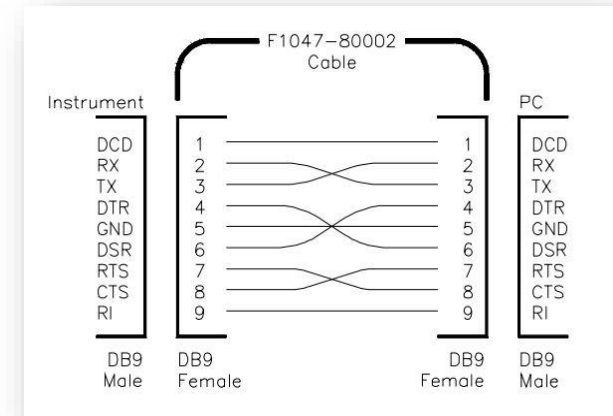
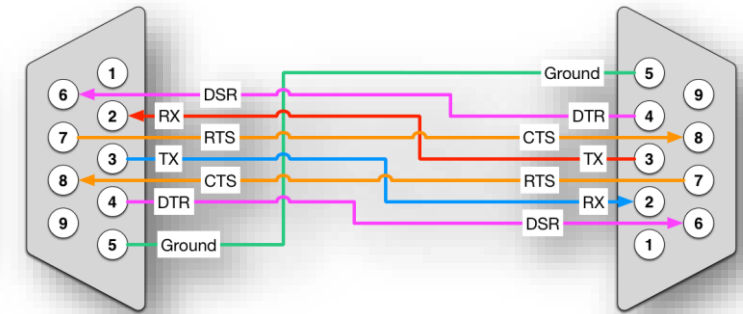
RS232 (Straight Through Cable)



RS232 (Cross-Over cable)



RS232 (Cross-Over with Handshake cable)





PROGRAM

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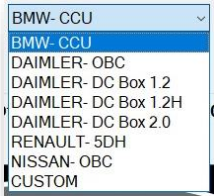


HOME



Test Program

- Choose program
- Fill serial number



Start testing

- When the test is running, DC source output is ON danger sign will blink.



Test Data

- Table contain list of contact Pairs and Maximum resistance from specification.

PE TESTING

File

Configuration

Help

Stop

Home

Edit Program

Manual

Database

Export Data

Exit



Test Program

Choose Program :

BMW- CCU

✓

Serial No. :

DES202012080061900041

Start testing (On/Off DC-Source Chroma 62000H)

Push foot button to Start ...

DANGER

Setpoint

Voltage :

5.00

V

Set

Current :

25.00

A

Set

Get data (Digital Multimeter - Keysight 34401a)

Test Result



mV

Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0	1.01	0.04	PASS
MP2 to MPE	10.0	1.05	0.04	PASS
MP3 to MPE	10.0	2.39	0.10	PASS
MP4 to MPE	10.0	3.48	0.14	PASS
MP1 to MPE (RE)	10.0	1.17	0.05	PASS
MP2 to MPE (RE)	10.0	1.08	0.04	PASS
MP3 to MPE (RE)	10.0	2.54	0.10	PASS
MP4 to MPE (RE)	10.0	3.24	0.13	PASS

Clear

Setpoint

- Set voltage and current to DC-source according to the test specification.

Test Result

- Voltage value from digital multimeter.

Test Data

- Measure voltage from digital multimeter.
- Calculated resistance.
- Result compared between maximum resistance and measured resistance.
- Clear button for clear the latest data that be listed in the table.

Edit Program



PE TESTING

File Configuration Help

Stop Home Edit Program Manual Database Export Data Exit

Edit Specification Test

Contact Pairs	Max.Res (mΩ)
Point A1 to Point B1	10
Point A2 to Point B1	10
Point A3 to Point B1	10
Point C1 to Point B2	20
Point C2 to Point B2	20

Insert

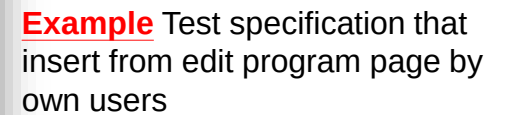
Clear

Ready

- Users can add new contact pair or new test specification.
- Test specification will import in the Test Data table.



Feature (Page and Panel)

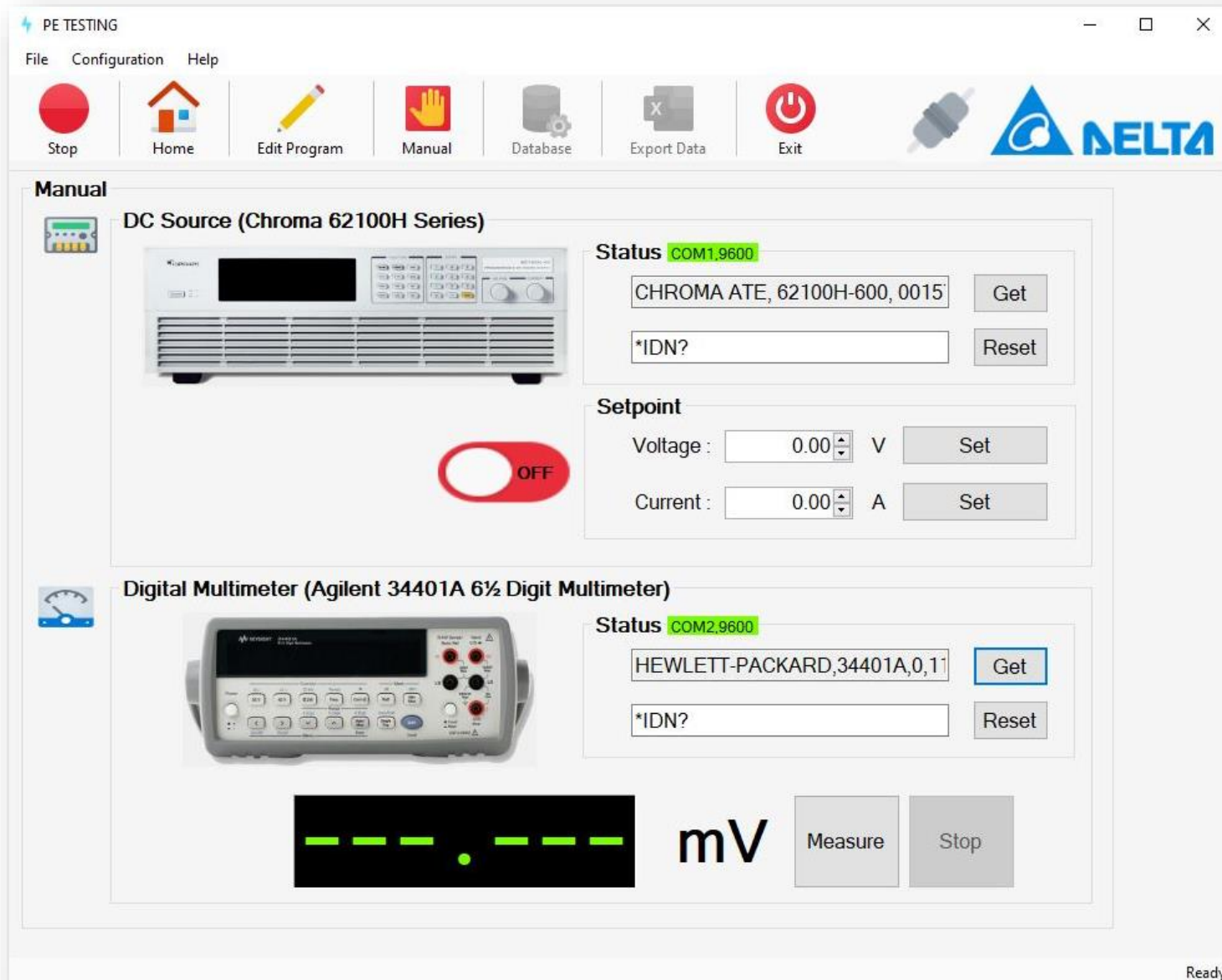


Manual



Shortcut Key : Ctrl + W

Feature (Page and Panel)



Manual page is contain equipment control that user can check status of the equipment, connection and the others control.

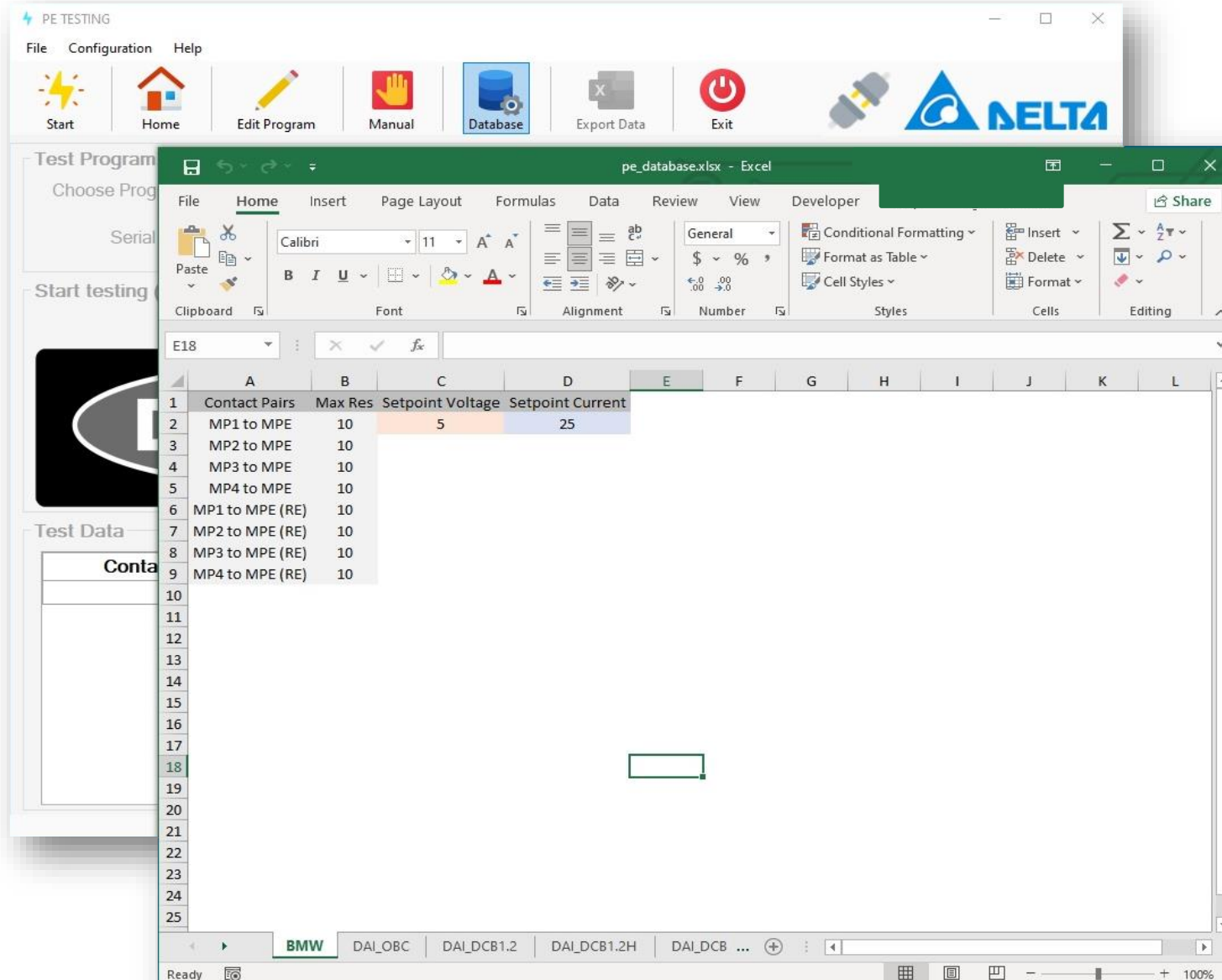


Database



Shortcut Key : Ctrl + R

Feature (Page and Panel)



- Users can click on **Database** icon to open database file that contain PE test specification in each project.
- Database is available to open, read and edit.
- Database file location **Drive D. > pe_database.xlsx**

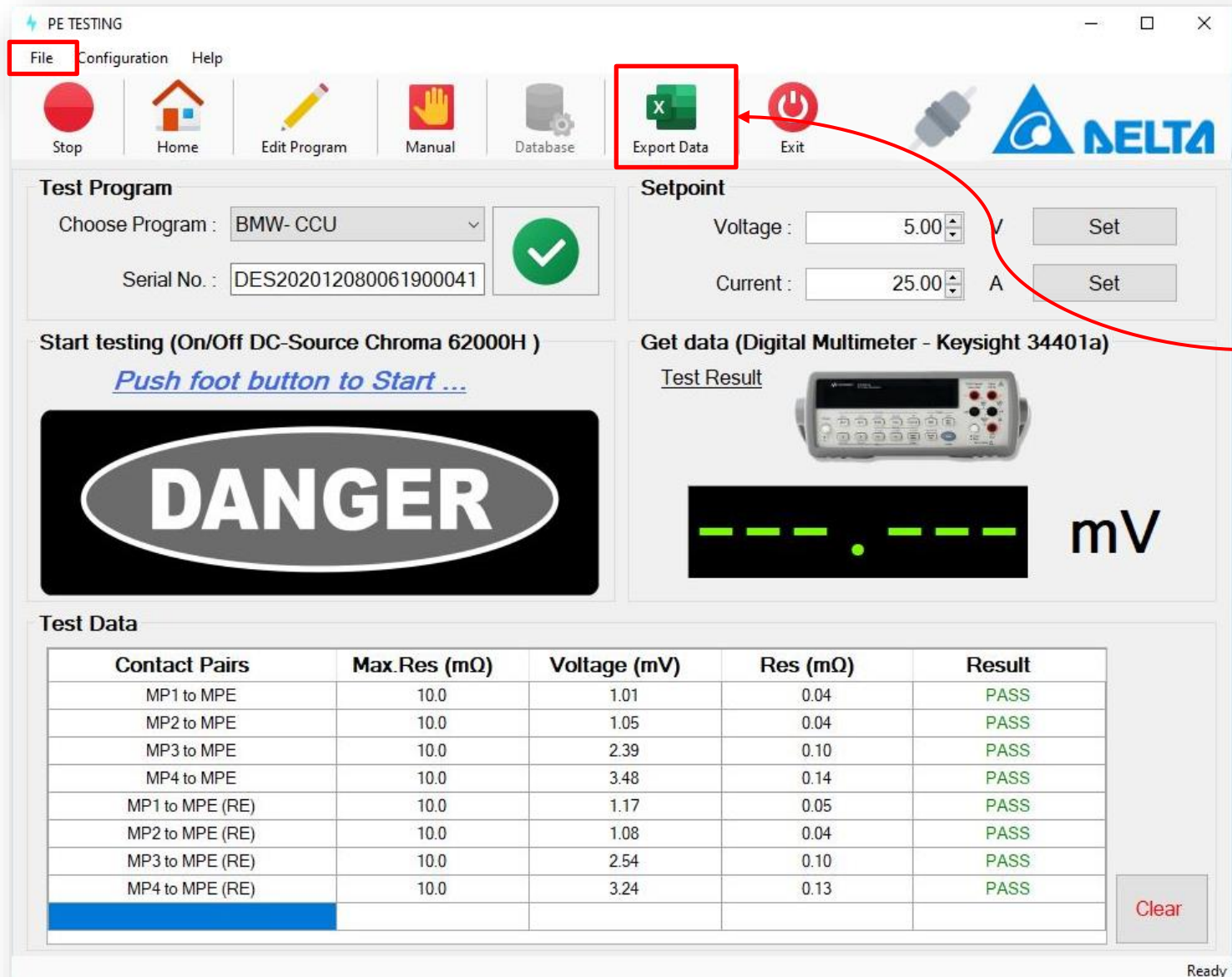


Microsoft Excel
Worksheet
Database file

Export Data



Shortcut Key : Ctrl + S



Test Program

Choose Program : BMW-CCU

Serial No. : DES202012080061900041

Start testing (On/Off DC-Source Chroma 62000H)

Push foot button to Start ...

DANGER

Setpoint

Voltage : 5.00 V

Current : 25.00 A

Get data (Digital Multimeter - Keysight 34401a)

Test Result

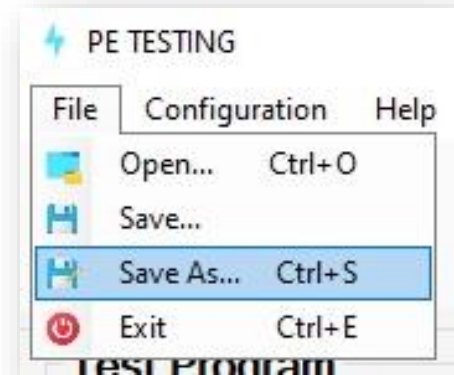
mV

Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0	1.01	0.04	PASS
MP2 to MPE	10.0	1.05	0.04	PASS
MP3 to MPE	10.0	2.39	0.10	PASS
MP4 to MPE	10.0	3.48	0.14	PASS
MP1 to MPE (RE)	10.0	1.17	0.05	PASS
MP2 to MPE (RE)	10.0	1.08	0.04	PASS
MP3 to MPE (RE)	10.0	2.54	0.10	PASS
MP4 to MPE (RE)	10.0	3.24	0.13	PASS

Clear

- When the testing is all done program will auto-generate the test report at default location **Drive D. > PE_DATA**
- users can export the report to any location by yourself by click **Export Data**. Or go to **File > Save As...**





Export Data



Feature (Page and Panel)

PE_SN0009_2021-09-29 14-49-21 - Excel													
File Home Insert Page Layout Formulas Data Review View Developer Tell me what you want to do...													
Clipboard Font Alignment Number Styles													
R1													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Project	DAIMLER- DC Box 1.2											
2	Serial No.	2200100021082600009											
3	Test Date	9/29/2021 14:49											
4	Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result								
5	MP1 to MPE	10	5.924	0.23696	PASS								
6	MP2 to MPE	10	23.046	0.92184	PASS								
7	MP3 to MPE	10	6.324	0.25296	PASS								
8	MP4 to MPE	10	9.272	0.37088	PASS								
9	MP5 to MPE	10	7.865	0.3146	PASS								
10	MP6 to MPE	10	7.908	0.31632	PASS								
11	MP7 to MPE	10	6.935	0.2774	PASS								
12	MP8 to MPE	10	6.625	0.265	PASS								
13	MP9 to MPE	10	2.165	0.0866	PASS								
14	MP1 to MPEC1	20	54.418	2.17672	PASS								
15	MP2 to MPEC1	20	71.652	2.86608	PASS								
16	MP3 to MPEC1	20	55.059	2.20236	PASS								
17	MP4 to MPEC1	20	57.856	2.31424	PASS								
18	MP5 to MPEC1	20	56.127	2.24508	PASS								
19	MP6 to MPEC1	20	56.105	2.2442	PASS								
20	MP7 to MPEC1	20	55.245	2.2098	PASS								
21	MP8 to MPEC1	20	54.924	2.19696	PASS								
22	MP9 to MPEC1	20	50.872	2.03488	PASS								
23	MPAC1 to MPEC1	20	59.993	2.39972	PASS								
24	MPAC2 to MPEC1	20	56.634	2.26536	PASS								
25	MP1 to MPE (RE)	10	5.885	0.2354	PASS								
26	MP2 to MPE (RE)	10	23.165	0.9266	PASS								
27	MP3 to MPE (RE)	10	6.423	0.25692	PASS								
28	MP4 to MPE (RE)	10	9.243	0.36972	PASS								
29	MP5 to MPE (RE)	10	7.92	0.3168	PASS								
30	MP6 to MPE (RE)	10	7.885	0.31544	PASS								

Example Excel file that exported contain Project name, Serial number, Test date and all test result.



Microsoft Excel
Worksheet

Example report

HOW TO ...

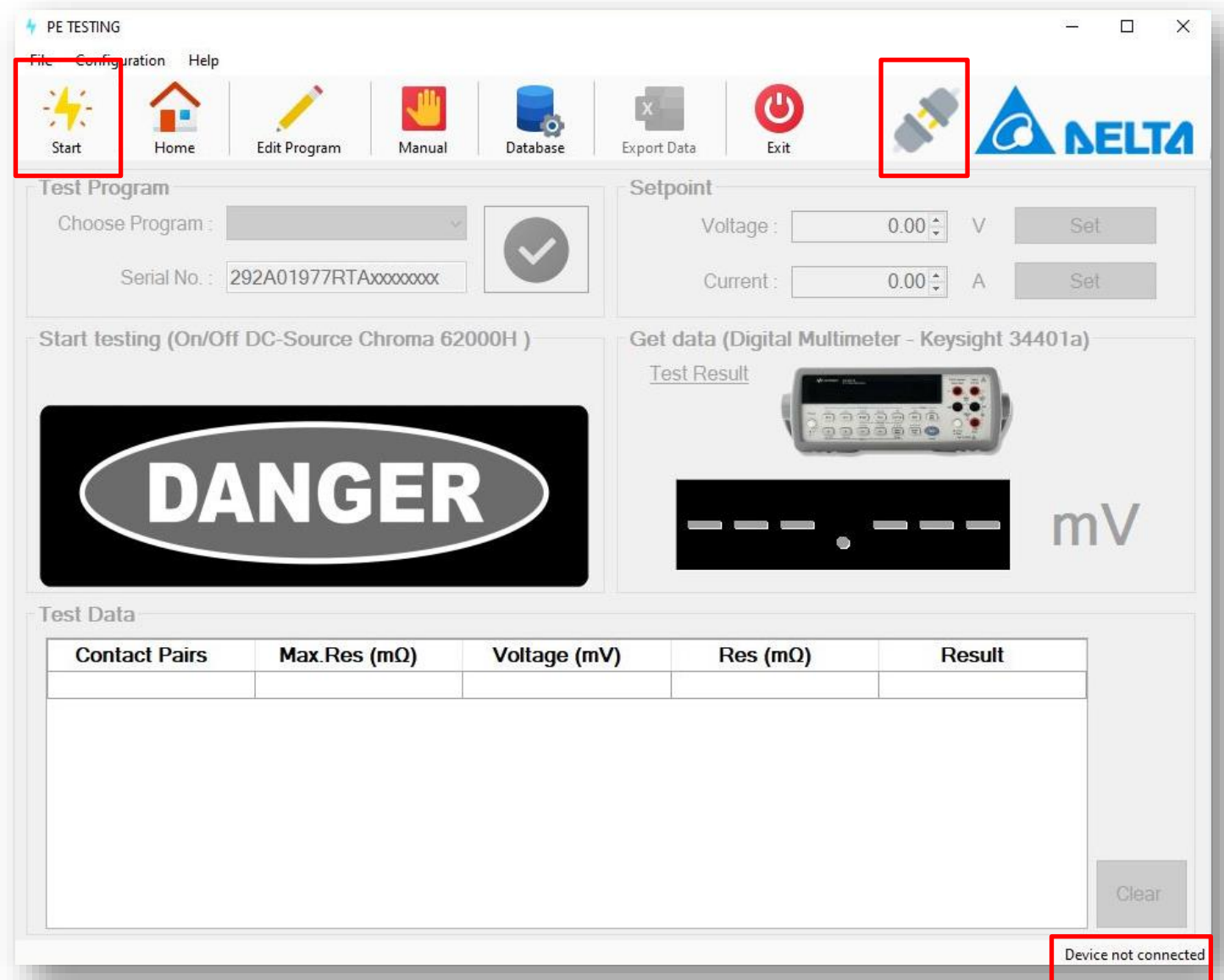
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How to use ?

1.Start



- Every equipment will be connected.
- Equipment will be commanded to access remote mode.
- All panel are enable.
- Program will enable to choose test program.
- Program status change to [Ready](#).
- Connection status turn to Connected.



How to use ?

1.Start

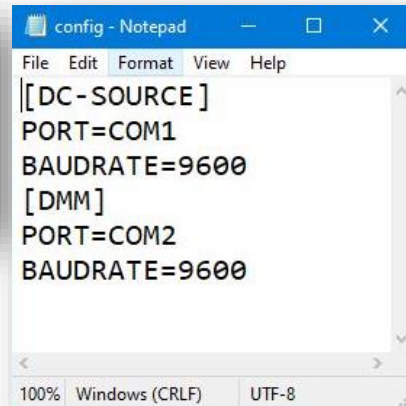
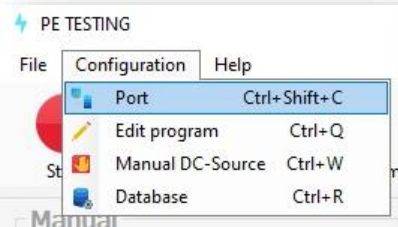


What if...

- If some panel not enable (become grey), It's mean equipment is not complete connection.

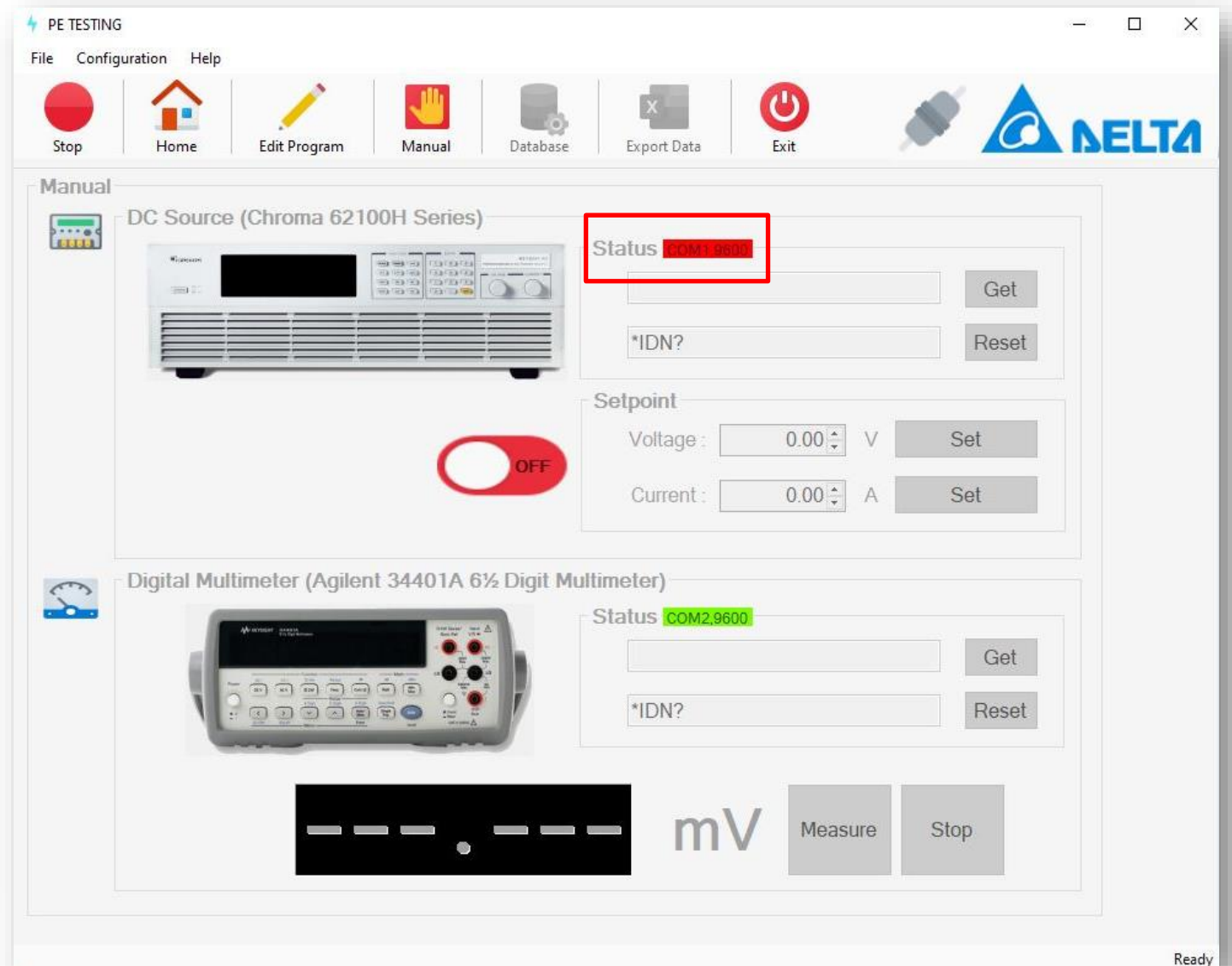
Solution :

1. Go to **Manual** and check at equipment status, the port that equipment is not connect become to **Red**.
2. Then go to verify the connection between PC and equipment, comport, equipment is on ?, equipment interface setting (RS232, Buadrate : 9600)
3. Go to **Configuration > Port (Ctrl+Shift+C)** to open the config.ini file that you can configuration the comport.
4. Config.ini file location **Drive D. > config.ini**
5. Click **Stop** and **Start** again



config.ini

Example config.ini

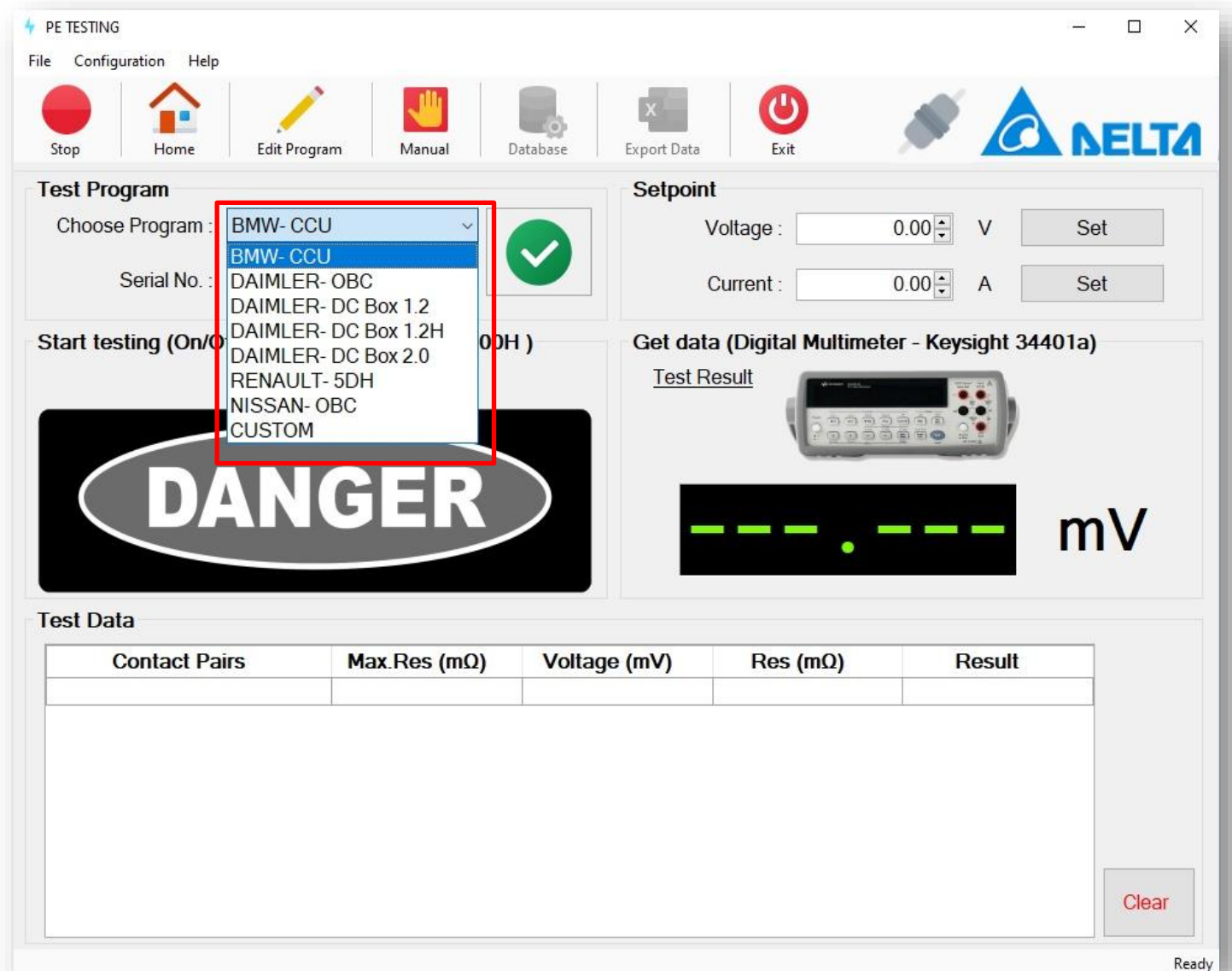




How to use ?

2.Choose Program

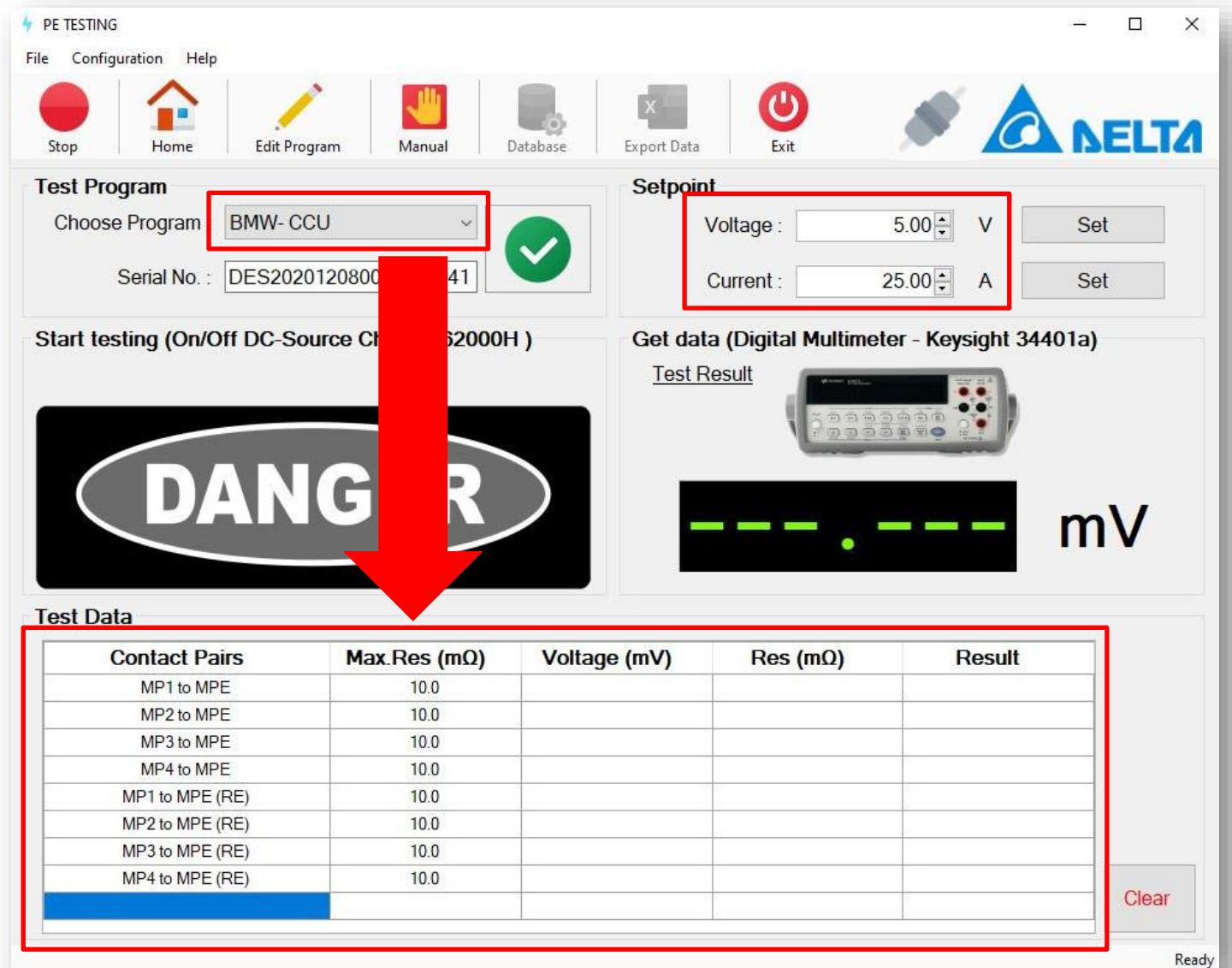
1. Choose program in drop down box.
2. Fill Serial number (maximum length 22 char.)
3. Click [correct button](#) to confirm the test program then wait for a second.



How to use ?

2.Choose Program

1. Choose program in drop down box.
2. Fill Serial number (maximum length 22 char.)
3. Click [correct button](#) to confirm the test program then wait for a second.
4. Then test specification will be listed on the Test Data table and voltage and current set point will be set to DC-Source automatically.



PE TESTING

File Configuration Help

Stop Home Edit Program Manual Database Export Data Exit

Test Program

Choose Program BMW-CCU 

Serial No. : DES202012080041

Setpoint

Voltage : 5.00 V Set

Current : 25.00 A Set

Start testing (On/Off DC-Source Ch 1 2000H)

DANGER

Get data (Digital Multimeter - Keysight 34401a)

Test Result

 mV

Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0			
MP2 to MPE	10.0			
MP3 to MPE	10.0			
MP4 to MPE	10.0			
MP1 to MPE (RE)	10.0			
MP2 to MPE (RE)	10.0			
MP3 to MPE (RE)	10.0			
MP4 to MPE (RE)	10.0			

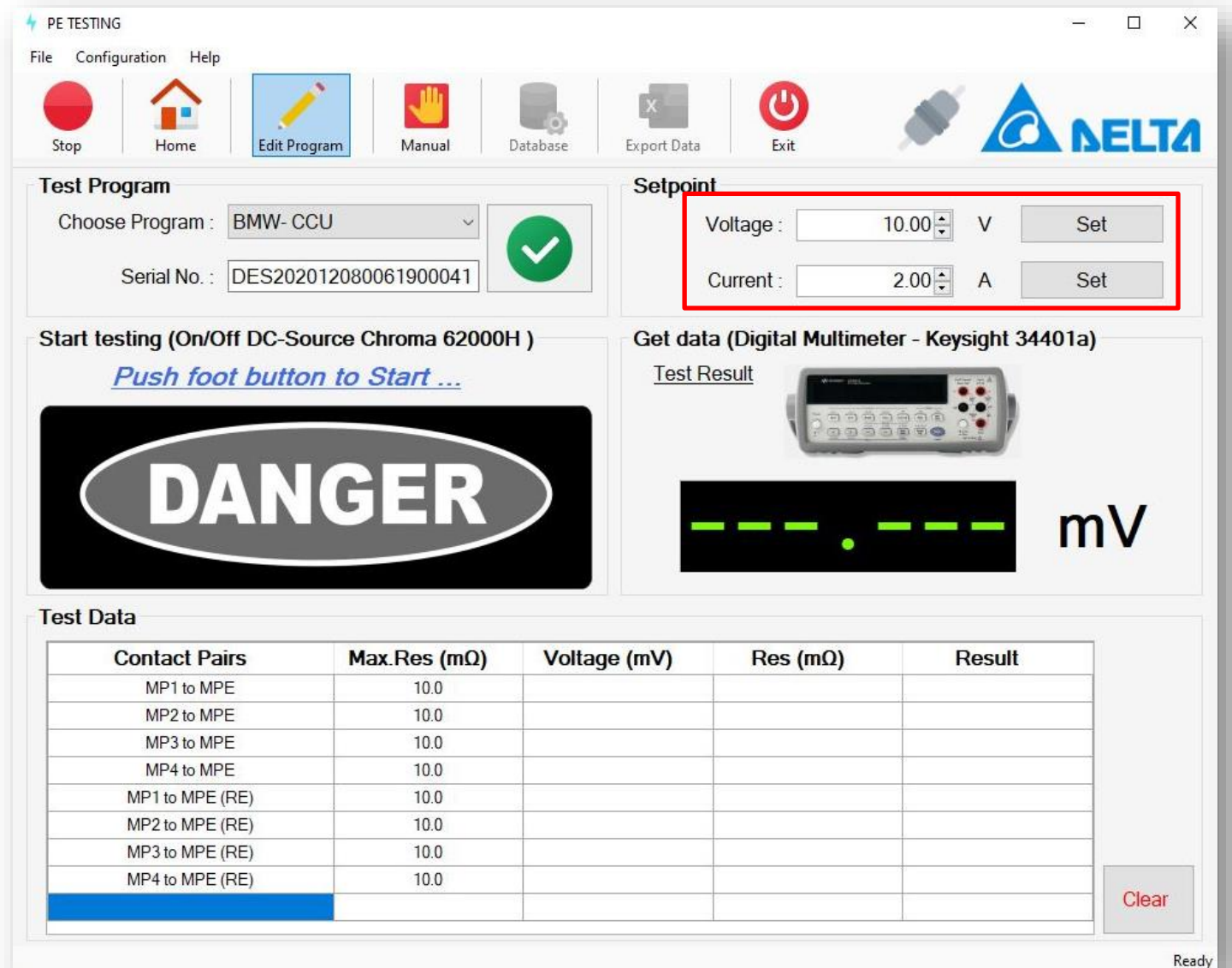
Clear

Ready

How to use ?

3.Set Point

- If you want to change voltage and current aside from specification defined you can change in up-down button (Increment 0.05) then [Enter](#) or click [Set](#) then program will send command to DC-Source.



The screenshot shows the PE TESTING software interface. The 'Setpoint' section is highlighted with a red box, showing the following controls:

- Voltage :** 10.00 V (with up/down arrows) and a **Set** button.
- Current :** 2.00 A (with up/down arrows) and a **Set** button.

Other visible sections include:

- Test Program:** Choose Program: BMW- CCU, Serial No.: DES202012080061900041, and a green checkmark button.
- Start testing (On/Off DC-Source Chroma 62000H)** with a blue link [Push foot button to Start ...](#) and a large **DANGER** warning sign.
- Get data (Digital Multimeter - Keysight 34401a)** showing a test result of **mV** with a green dashed line and a dot.
- Test Data** table with columns: Contact Pairs, Max.Res (mΩ), Voltage (mV), Res (mΩ), and Result.
- Clear** button at the bottom right.
- Ready** status at the bottom right.

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0			
MP2 to MPE	10.0			
MP3 to MPE	10.0			
MP4 to MPE	10.0			
MP1 to MPE (RE)	10.0			
MP2 to MPE (RE)	10.0			
MP3 to MPE (RE)	10.0			
MP4 to MPE (RE)	10.0			

How to use ?

3. Testing



1. Push the couple of electrical probe to sample.
2. Push on the foot switch - output of DC-Source is turn to **ON**, digital mutimeter start measure, danger sign will blink, Test program, Setpoint and Manual panel disable to protect some human error while the output of DC-Source is ON and the program status change to Testing...
3. When the voltage is stable you can push on the foot switch again - output of DC-Source is turn to **OFF**, DMM is stop measure, danger sign stop blinking and the value and result will appears on the table.

Note: Program use voltage value to calculate to be resistance and compare with maximum resistance, if resistance from measured not more than maximum resistance from specification the result will be **PASS** then else the result become **FAIL**

Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0	37.84	1.51	PASS
MP2 to MPE	10.0	33.24	1.33	PASS
MP3 to MPE	10.0	24.13	0.97	PASS
MP4 to MPE	10.0	30.13	1.21	PASS
MP1 to MPE (RE)	10.0	261.32	10.45	FAIL
MP2 to MPE (RE)	10.0			
MP3 to MPE (RE)	10.0			
MP4 to MPE (RE)	10.0			

PE TESTING (RUNNING)

File

Configuration

Help

Stop

Home

Edit Program

Manual

Database

Export Data

Exit



Test Program

Choose Program :

BMW-CCU

Serial No. :

DES202012080061900041

Setpoint

Voltage :

5.00

V

Set

Current :

25.00

A

Set

Start testing (On/Off DC-Source Chroma 62000H)

Push foot button to Stop ...



Get data (Digital Multimeter - Keysight 34401a)

Test Result



37.840

mV

Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0			
MP2 to MPE	10.0			
MP3 to MPE	10.0			
MP4 to MPE	10.0			
MP1 to MPE (RE)	10.0			
MP2 to MPE (RE)	10.0			
MP3 to MPE (RE)	10.0			
MP4 to MPE (RE)	10.0			

Clear

Testing...

How to use ?

3. Testing

What if...

- If you want to verify and re-measure on the last contact pairs again you can click at Clear button on the right side of table then the last data will be deleted.



Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0	37.84	1.51	PASS
MP2 to MPE	10.0	33.24	1.33	PASS
MP3 to MPE	10.0	24.13	0.97	PASS
MP4 to MPE	10.0	30.13	1.21	PASS
MP1 to MPE (RE)	10.0			
MP2 to MPE (RE)	10.0			
MP3 to MPE (RE)	10.0			
MP4 to MPE (RE)	10.0			

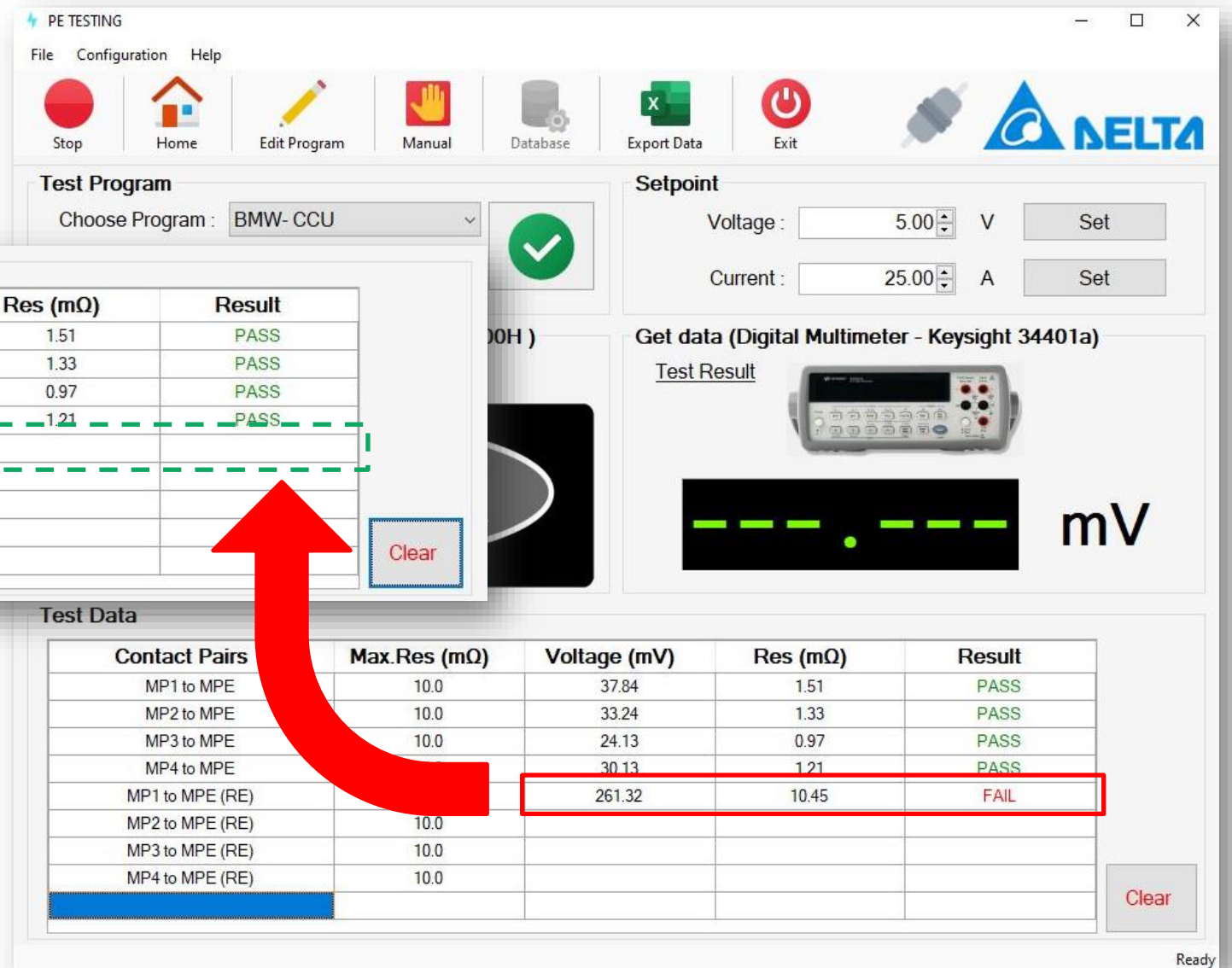
Clear

Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0	37.84	1.51	PASS
MP2 to MPE	10.0	33.24	1.33	PASS
MP3 to MPE	10.0	24.13	0.97	PASS
MP4 to MPE	10.0	30.13	1.21	PASS
MP1 to MPE (RE)	10.0	261.32	10.45	FAIL
MP2 to MPE (RE)	10.0			
MP3 to MPE (RE)	10.0			
MP4 to MPE (RE)	10.0			

Clear

Ready



How to use ?

3. Testing



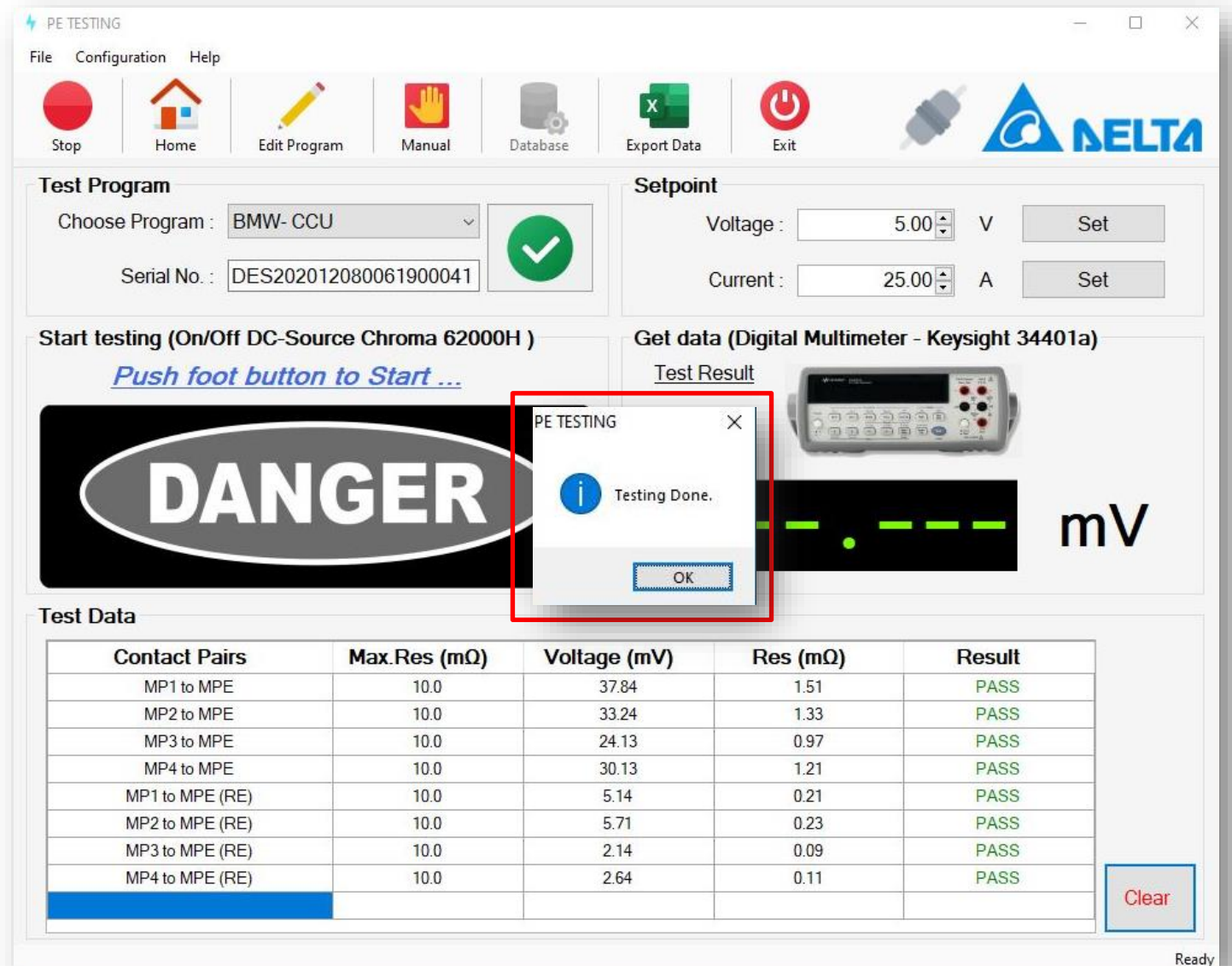
What if...

- When your test data is complete but you push the foot switch accidentally make DC-Source output is on and the measure is start, you can push the foot switch again to stop testing after that program will pop-up the warning box that inform you "Done !!!" then click OK and click [Stop](#)



What if... (Update)

- When your last point test data is complete, after you push foot switch to turn off DC-source – Popup box (Testing Done.) will appears immediately then you can click OK and Go to stop program.



The screenshot shows the PE TESTING software interface. At the top, there's a menu bar with File, Configuration, and Help. Below it is a toolbar with icons for Stop, Home, Edit Program, Manual, Database, Export Data, and Exit. The main area is divided into several sections:

- Test Program:** Includes a dropdown for 'Choose Program' (set to BMW-CCU) and a 'Serial No.' field (DES202012080061900041). A green checkmark icon is visible.
- Setpoint:** Includes fields for 'Voltage' (5.00 V) and 'Current' (25.00 A), each with a 'Set' button.
- Start testing (On/Off DC-Source Chroma 62000H):** Includes a button labeled 'Push foot button to Start ...'.
- Get data (Digital Multimeter - Keysight 34401a):** Includes a 'Test Result' section with a digital multimeter image and a large 'mV' display.
- Test Data:** A table showing test results for various contact pairs.

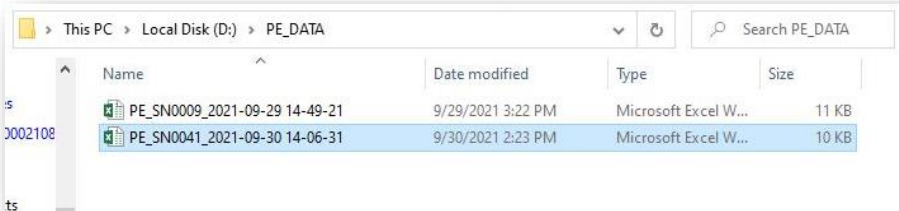
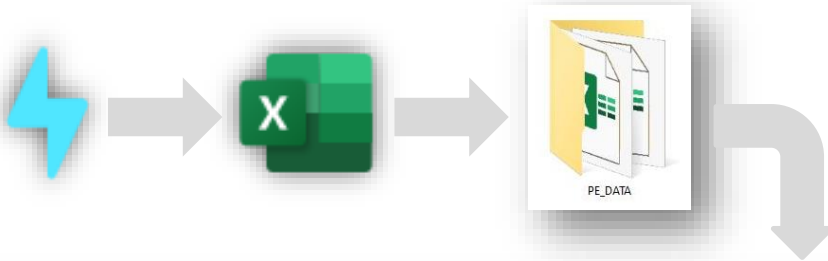
A red box highlights a 'PE TESTING' popup window with the message 'Testing Done.' and an 'OK' button. A large 'DANGER' sign is also visible in the background of the interface.

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0	37.84	1.51	PASS
MP2 to MPE	10.0	33.24	1.33	PASS
MP3 to MPE	10.0	24.13	0.97	PASS
MP4 to MPE	10.0	30.13	1.21	PASS
MP1 to MPE (RE)	10.0	5.14	0.21	PASS
MP2 to MPE (RE)	10.0	5.71	0.23	PASS
MP3 to MPE (RE)	10.0	2.14	0.09	PASS
MP4 to MPE (RE)	10.0	2.64	0.11	PASS

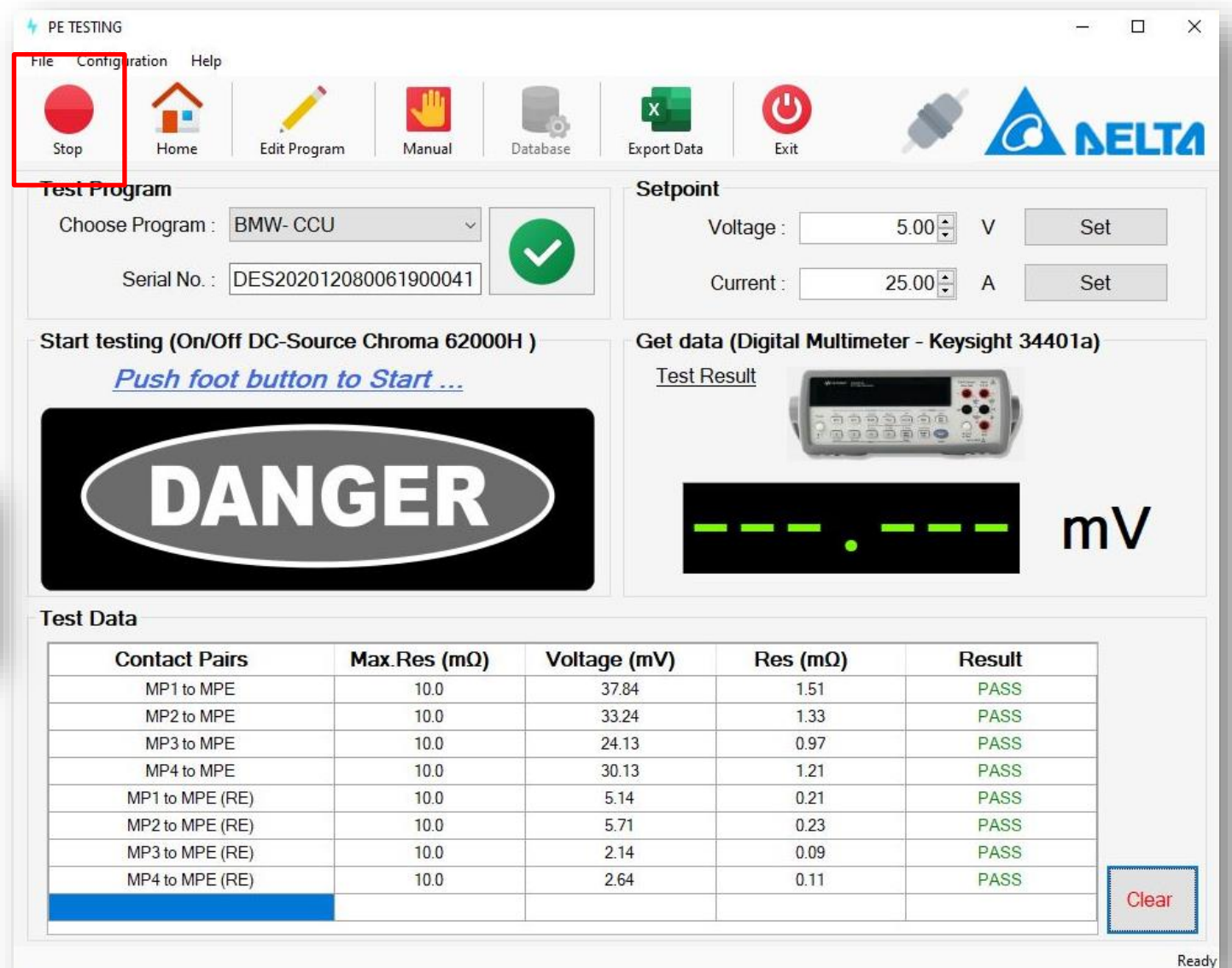
At the bottom right, there is a 'Clear' button and a 'Ready' status indicator.

3. Testing

- When your test is finished you can click **Stop** then the report is auto-generate at the default location **Drive D. > PE_DATA**.



How to use ?



PE TESTING

File Configuration Help

Stop Home Edit Program Manual Database Export Data Exit

Test Program

Choose Program : BMW-CCU

Serial No. : DES202012080061900041

Setpoint

Voltage : 5.00 V

Current : 25.00 A

Start testing (On/Off DC-Source Chroma 62000H)

Push foot button to Start ...

DANGER

Get data (Digital Multimeter - Keysight 34401a)

Test Result

mV

Test Data

Contact Pairs	Max.Res (mΩ)	Voltage (mV)	Res (mΩ)	Result
MP1 to MPE	10.0	37.84	1.51	PASS
MP2 to MPE	10.0	33.24	1.33	PASS
MP3 to MPE	10.0	24.13	0.97	PASS
MP4 to MPE	10.0	30.13	1.21	PASS
MP1 to MPE (RE)	10.0	5.14	0.21	PASS
MP2 to MPE (RE)	10.0	5.71	0.23	PASS
MP3 to MPE (RE)	10.0	2.14	0.09	PASS
MP4 to MPE (RE)	10.0	2.64	0.11	PASS

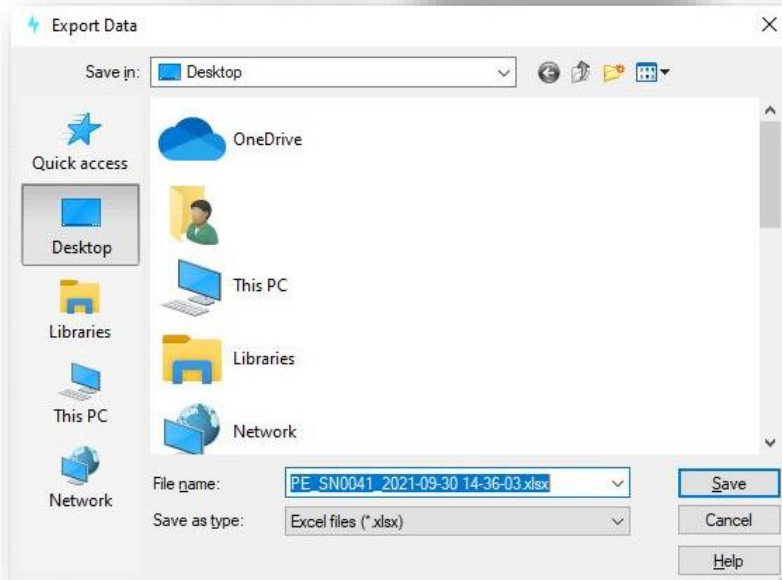
Ready

4.Export

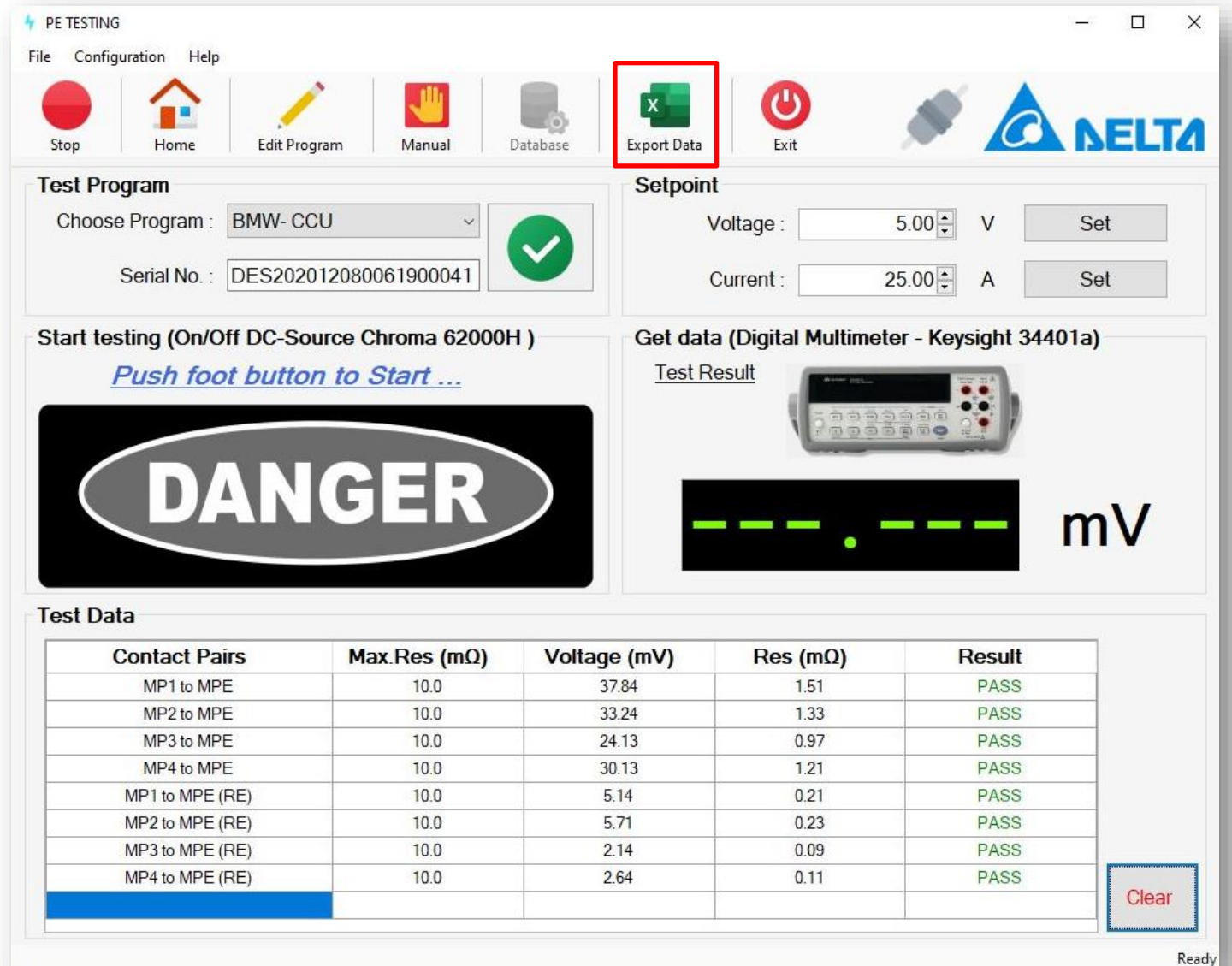


What if...

- If you want to generate report to with the other file name or the other location, Before **Stop** you can click **Export Data** button or go to **File > Save As...** after that Export Data box will pop-up for you to change prefer name and file location then click Save.



How to use ?



Additional

[**Back to content**](#)

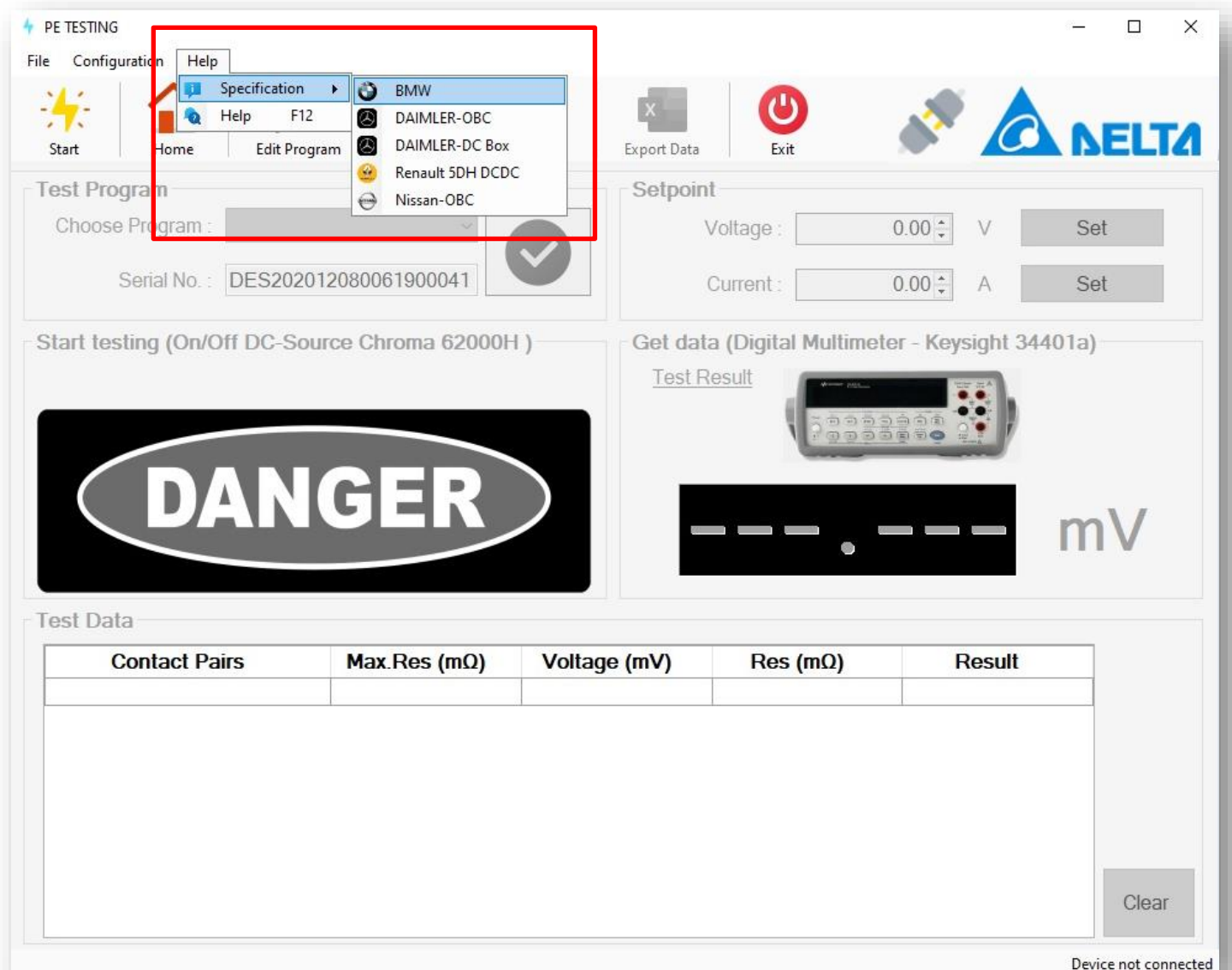


Additional

Help



- This program have information of each project specification you can go to [Help > Specification > Project \(BMW and etc.\)](#)



Device not connected



Help



- This program have information of each project specification you can go to [Help > Specification > Project \(BMW and etc.\)](#)



BMW 22kW and BMW 7.4kW Specification

Additional

BMW Specification

OK

DELTA

PE Test

EQUIPMENT :

1. DC Source	: Chroma 62100H-40
2. Digital Multimeter	: Keysight 34461A
3. Voltage probe	: Burster Kelvin Probes Type 2387-V001

TEST SPEC :

☐ Customer Spec. ☒ DES Soest ☐ QE Test Procedure

☐ Other Standard : BMW_CCU_22KW_PETest_02_ECD180200xx

TEST PARAMETER :

Ambient Temperature	: Room Temperature 23°C ±5K
Humidity	: uncontrolled
Current	: 25A + 0,5A
U _{max}	: ≤ 5V

Picture 1: MPE - MP1

PE contact: MPE

Picture 2: MP2

MP2

MP1

MP3



Help



- This program have information of each project specification you can go to [Help > Specification > Project \(BMW and etc.\)](#)



DC Box 2.0 and DC Box 1.2 Specification

Additional

DAIMLER-DC Box Specification

OK

DELTA

PE Test

EQUIPMENT :

1. DC Source	: Chroma 62100H-40
2. Digital Multimeter	: Keysight 34461A
3. Voltage probe	: Burster Kelvin Probes Type 2387-V001

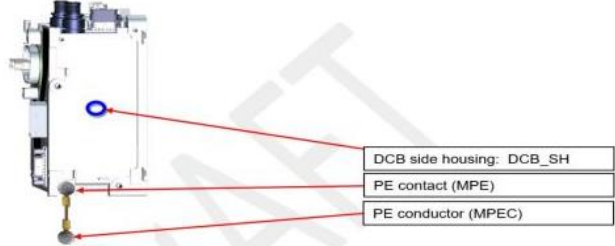
TEST SPEC :

☐ Customer Spec.	☒ DES Soest	☐ QE Test Procedure
☐ Other Standard :	2018_04_19_QTS_PETest_ECD90010012_Rev00.2	

TEST PARAMETER :

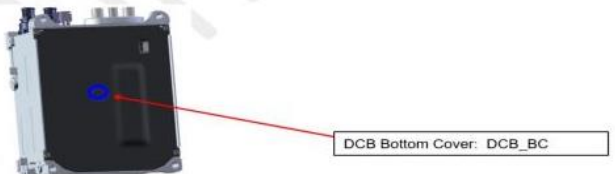
Ambient temperature: Room Temperature 23°C ±5K
Humidity: uncontrolled
Current: 25A
Umax: >5V (<10V)

Picture 1: DCBox side housing vs. DCBox PE connections



DCB side housing: DCB_SH
PE contact (MPE)
PE conductor (MPEC)

Picture 2: DCBox bottom housing vs. DCBox PE connections



DCB Bottom Cover: DCB_BC



Help



- This program have information of each project specification you can go to [Help > Specification > Project \(BMW and etc.\)](#)



Renault 5DH Specification

Additional

Renault 5DH Specification

OK

DELTA

PE Test

EQUIPMENT :

1. DC Source	: Chroma 62100H-40 [5V/25A]
2. Digital Multimeter	: Keysight 34461A
3. Voltage probe	: Burster Kelvin Probes Type 2387-V001

TEST SPEC :

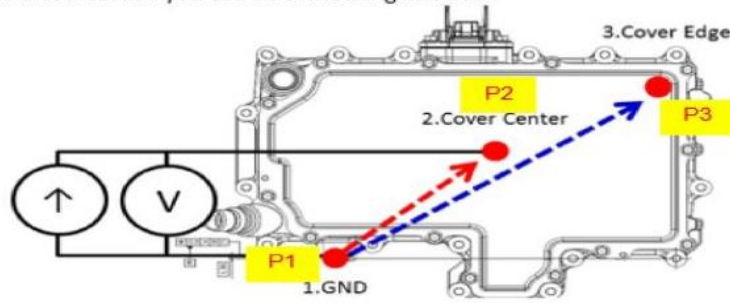
☒ Customer Spec. ☐ DES Soest ☐ QE Test Procedure

☐ Other Standard : 20200605_Renault-5DH-DCDC_ENV_TS_Safety_Rev04.pdf

TEST PARAMETER :

Ambient Temperature	: Room Temperature 23°C ±5K
Humidity	: uncontrolled
Current	: 1A
U_{max}	: 20mV

For electrical risk you can do something like this :



ACCEPTANCE CRITERIA :

The measured voltage in both case must be less than 20mV ($20mV/1A = 20m\Omega$)

GND (P1) to Cover Center (P2)

Result

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- Easy test more than manual test
- Take the less time for testing. Ex. DC-Box 1.2 that have the most testing point .
 (Manual test) : about 20 minutes).
 (Semi-auto test) : about 10 minutes.
- Reduced electric arc because tester take the less time while push the electric probe on sample.
- Not to use more tester to record the value.
- Easy to keep data.

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